Decision AI Model (Cloudflare Clef)

AGENT KIO • STUDENT 2 RESPONSIBILITIES & IMPLEMENTATION REPORT

1. Executive Summary

We built an instruction-based decision router utilizing the **Cloudflare Workers AI — Clef** model. By supplying structured routing guidelines to Clef via an authenticated API, the system evaluates incoming customer messages without training a traditional ML classifier. On the frozen 20-message test set, Clef achieved **100% accuracy (20/20)** with an average latency of ~1,467 ms.

2. Problem Definition & Intent Rules

Category	Flag	Scope & Routing Criteria
Support	**0**	Problems, complaints, billing issues, account access, existing service repairs.
Sales	**1**	New purchases, package upgrades, price quotes, promotions, new plans.

Mixed-Intent & Negation Rules:

- **Mixed Intent:** If a message contains both an existing problem and interest in buying, **Support (Flag 0) takes priority**.
- **Negation Handling:** The model must parse contextual intent rather than matching keywords (e.g., declining a purchase while asking for a fix → Support).

3. Dataset & Evaluation Integrity

Evaluated on `Agent_KIO_Dataset_FROZEN.xlsx` (60 total synthetic messages: 40 Train / 20 Test). Paraphrase groups (`Group_ID`) do not cross splits, and dataset integrity is locked via SHA-256 content hash. The decision model receives only instructions and message text—it is never trained on the data or shown correct labels.

4. Approach: Instruction-Based Decision AI

Instead of relying on statistical term frequency, Clef processes structured Markdown rules describing class definitions, mixed-intent priority, negation rules, and output schemas. Inputs consist of: (1) `routing_instructions.md`, (2) user query, and (3) binary choice specification (`support` vs. `sales`).

5. Provider Setup & API Configuration

Item	Details
Provider & Model	Cloudflare Workers AI — Clef (`@cf/cloudflare/clef`)
Integration Type	Hosted API model; zero-shot inference without fine-tuning
License & Terms	**[FILL IN from Cloudflare's model page & terms]**
Authentication	Cloudflare Account ID & API Token stored in local `.env`

* Provider Selection Note: Switched to Cloudflare Workers AI after the primary option lacked available API credits. Credentials remain fully isolated from version control.

6. Code Architecture & Modules

File	Responsibility
`kio/clef_router.py`	Dispatches requests to Cloudflare API and parses returned payload.
`kio/validation.py`	Validates output labels (`support` / `sales`) and maps to flags (0 / 1).
`kio/routing_instructions.md`	Defines explicit system instructions and decision guidelines.
`decision_model/evaluate_decision_model.py`	Executes evaluation pipeline across the 20 frozen test messages.

7. Output Validation & Error Guardrails

To prevent bad predictions on system errors, failures instantly trigger an **invalid** status rather than a guessed label. Triggers include missing API tokens, rate limits, timeouts, HTTP errors, malformed JSON, or probability values out of range. Valid labels strictly map `support` → 0 and `sales` → 1.

8. Test Set Performance (20 Frozen Messages)

Metric	Clef Result	Metric	Clef Result
Correct / Total	**20 / 20**	**Sales Precision**	1.000 (100%)
Accuracy	**100%**	**Sales Recall / F1**	1.000 / 1.000
Average Latency	~1,467 ms	**Confidence Range**	0.90 to 0.99

9. Diagnostic Analysis & Success Cases

Clef successfully resolved nuanced phrasing, complex paraphrases, and both negation edge cases that caused the Traditional ML model to fail:

- **ID 59:** *"I am not interested in TV, just give me the internet prices."* → **Sales (1)** [Correct]
- **ID 60:** *"No TV channels, just pure fiber broadband, how much?"* → **Sales (1)** [Correct]

This confirms that instruction-following LLMs parse syntactic relationship structures rather than relying on term co-occurrence.

10. Comparative Evaluation: Decision AI vs. Traditional ML

Dimension	Cloudflare Clef (Decision AI)	Traditional ML (TF-IDF + LogReg)
Accuracy (20 Test Messages)	**100% (20/20)**	90% (18/20)
Average Latency	~1,467 ms	**~8.8 ms** (~170x faster)
Confidence Output	High (0.90 – 0.99)	Low (~0.50 – 0.52)
Training Data Needed	None (Zero-shot)	40 labelled examples
Infrastructure / Cost	Cloud API & Internet required	Runs locally offline; zero cost

11. Limitations & Architectural Recommendation

- **Sample Size Constraint:** 100% on 20 test messages shows strong prompt alignment but does not guarantee performance on noisy real-world data.
- **Latency & External Risk:** Clef introduces external network dependencies and higher response latency (~1.47 seconds per query).
- **Recommendation:** Adopt a **Hybrid Router Model**—route incoming requests to the ultra-fast local ML model first, falling back to Clef when ML confidence is low or negation keywords are present.

12. Sources & Environment Setup

• Cloudflare Workers AI Docs & Clef Page: **[ADD URL]**
• Cloudflare Terms & License: **[ADD URL]**
• Dependencies: Python 3.14, pandas, scikit-learn, requests, python-dotenv, Streamlit